

EXPERIENCE

Database Engineer

November 2025 – present

Keyanna Technology, Ahmedabad

- Managed and maintained **24+ production MySQL databases** across multiple business-critical applications and **3 international regions**, ensuring high availability, operational stability, and performance optimization.
- Administered **MySQL Replication, InnoDB Cluster, DC/DR environments, failover operations, and recovery procedures**, supporting fault-tolerant and highly available database infrastructure.
- Diagnosed and resolved critical production incidents involving **replication lag, split-brain scenarios, deadlocks, locking contention, storage exhaustion, failed imports/exports, and resource bottlenecks** on Linux-based database servers.
- Performed advanced **query optimization, index tuning, execution plan analysis, and slow query remediation** using tools such as **pt-query-digest**, improving performance of high-volume transactional workloads.
- Designed and implemented **automated backup, restore, and disaster recovery workflows** using Shell scripting and Cron-based scheduling, improving operational reliability and recovery readiness.
- Built and maintained multiple **database automation processes** for routine operational activities, monitoring, maintenance, and reporting, significantly reducing manual intervention in production environments.
- Led large-scale **schema enforcement and data integrity initiatives** on production datasets exceeding **100M+ records**, implementing business-rule-driven uniqueness validation while preserving historical data and ensuring zero disruption to application functionality.
- Designed and executed **partitioning strategies for large transactional datasets** in clustered MySQL environments, successfully maintaining cluster stability during deployment and ongoing production operations.
- Performed **database growth analysis, capacity planning, and storage optimization**, identifying abnormal growth patterns and implementing long-term remediation strategies to improve infrastructure reliability.
- Executed **MongoDB to MySQL and MongoDB to PostgreSQL** migration projects involving schema transformation, data reconciliation, validation, and integrity verification.
- Developed **Grafana-based monitoring and visualization solutions**, creating optimized dashboards and analytical reports for clients, management, and CTO-level infrastructure reviews.
- Acted as a primary contributor during high-severity production incidents, performing root-cause analysis, recovery planning, and preventive engineering measures to improve database platform resilience and operational continuity.

Research Intern

May 2025 – July 2025

IIST, Shibpur

- Engineered a lightweight, noise-robust algorithm for **audio fingerprint extraction**, achieving **98.78% accuracy** even under signal distortion.
- Validated algorithm on **3+ large-scale datasets**, demonstrating **98.7% deployability success** on resource-constrained hardware and supporting real-world adoption.

Full Stack Developer

June 2024 – August 2024

LernTricks (Remote) [Certificate]

- Led full-cycle development and launch of a web platform, streamlining admin workflows and increasing efficiency by **60%**.
- Designed scalable data models and authentication flows using **Mongoose, Node.js, and Express.js**, reducing server response time by **35%**.

CERTIFICATIONS

- **Database Programming with SQL | Oracle** | September 2023
- **NDG Linux Essentials | Cisco Networking Academy** | May 2024
- **AWS Academy Data Engineering | AWS Academy** | October 2024

PROJECTS & RESEARCH IMPLEMENTATIONS

- Object Detection using XAI for Automated Vehicles

June 2025 – August 2025

 - Developed an object detection system for autonomous vehicle scenarios, integrating explainable AI techniques for transparent hazard identification.
 - Designed data processing and model evaluation workflows using large-scale image datasets, improving interpretability across **90%** of test cases.
- Hybrid Audio Fingerprinting of Human Voice Under Distortion

May 2025 – July 2025

 - Engineered a noise-robust audio fingerprinting algorithm achieving **98.78% accuracy** under distorted signal conditions.
 - Built data ingestion, preprocessing, and validation pipelines to evaluate performance across multiple large-scale datasets.
- AI-Driven Agricultural E-Commerce Platform (MERN)

Jan 2024 – July 2024

 - Developed a full-stack MERN application supporting user management, product catalog management, transaction workflows, and scalable data storage.
 - Designed database schemas and optimized backend data access patterns to improve application performance and reliability.
 - Published a conference paper on AI-assisted crop recommendation using soil and seasonal data analysis.
- Deepfake Detection Model

July 2024 – Sept 2024

 - Developed a deepfake detection solution achieving **94.84% accuracy** on benchmark datasets using deep learning techniques.
 - Integrated the solution into a web-based platform for real-time image validation and automated prediction workflows.

TECHNICAL SKILLS

- Database Administration:** MySQL, PostgreSQL (Basic), Replication, InnoDB Cluster, Backup & Recovery, Disaster Recovery (DC/DR), High Availability (HA), Failover Management, Query Optimization, Indexing, Partitioning, Performance Tuning, Capacity Planning, Database Migration
- SQL & Data Management:** SQL, Query Analysis, Execution Plans, Data Validation, Data Integrity, Schema Design, Data Modeling, Stored Procedures, Transaction Management
- Automation & Scripting:** Shell Scripting, Cron Jobs, Python Automation, Linux Administration, Process Automation
- Monitoring & Observability:** Grafana Dashboards, Database Monitoring, Metrics Analysis, Capacity Analysis, Root Cause Analysis, Performance Monitoring
- Programming Languages:** SQL, Python, JavaScript, Java, C++
- Cloud, DevOps & Tools:** Docker, Git/GitHub, Postman, Tableau
- Web Technologies:** MongoDB, Node.js, Express.js, React.js, Next.js, Spring Boot
- Machine Learning & Data Science:** Scikit-Learn, TensorFlow, Keras, Pandas, NumPy, OpenCV, Model Evaluation

ACHIEVEMENTS LEADERSHIP

- Production Database Engineering:** Successfully implemented business-rule-driven uniqueness enforcement on production datasets exceeding 100M+ records while preserving historical data integrity and application continuity.
- High Availability Operations:** Contributed to the stability and recovery of mission-critical MySQL environments involving replication, InnoDB Cluster, DC/DR operations, and production incident management.
- Postman X Marwadi University API Challenge:** Ranked **7th among 200+ participants**, demonstrating strong problem-solving skills in API integration, backend systems, and performance optimization.
- GeekForGeeks X VULTR Hackathon:** Ranked among the **Top 20 teams nationwide** for designing and deploying an AI-driven crop recommendation platform on cloud infrastructure.
- IEEE Region 10 Website Management:** Managed and maintained the official IEEE Region 10 website for over **2 years**, supporting content operations, platform reliability, and collaboration with regional teams.

EDUCATION